

SAM-Based Automotive Supply-Chain Attribution

This note documents how the SAM-derived outputs (especially ``data/intermediate/sam_auto_naics_shares.csv``) are generated from the raw social accounting matrix (``data/raw/SAM.csv``).

Source Data

- **Social Accounting Matrix** - ``data/raw/SAM.csv``
 - Fields: ``PayingCode``, ``PayingDescription``, ``ReceivingCode``, ``ReceivingDescription``, ``TransferCode``, ``TransferDescription``, ``Value``
 - Key transfer types:
 - ``Commodity Use``: intermediate demand by industries for commodities
 - ``Commodity Make``: commodity output supplied by producing industries
- **BEA Make Table** - ``data/raw/IOMake_Before_Redefinitions_PRO_Detail.xlsx`` (sheet ``2017``) to link SAM/BEA industries to NAICS prefixes
- **Mobility supply-chain lookup** - ``data/lookups/segment_assignments.csv`` containing the 38 target NAICS industries and segment metadata

Processing Workflow

Implemented in ``scripts/analyze_sam_auto_supply_chain.py``.

1. **Load and clean the SAM**
 - Strip whitespace from column names and coerce ``Value`` to numeric.
2. **Identify automotive purchasing industries**
 - Default list: SAM industries 324-336 (motor vehicles, bodies, transmissions, interiors, etc.). The list is overrideable through the ``--auto-codes`` argument.
3. **Aggregate commodity demand**
 - Filter ``Commodity Use`` rows.
 - For each commodity (``ReceivingCode``), compute total demand across all industries and the subset purchased by the automotive list.
 - Derive ``auto_share = auto_demand / total_demand``.
 - Output: ``sam_auto_commodity_shares.csv``.
4. **Allocate commodity demand to producing industries**
 - Filter ``Commodity Make`` rows.
 - For each commodity (``PayingCode``), compute the share supplied by each industry (``make_share = industry value / total commodity output``).
 - Multiply ``make_share`` by the commodity-level ``auto_demand`` to obtain the portion of each industry's output sold to automotive buyers.
 - Sum by industry to get ``auto_attributed_output`` and retain each industry's total output (``total_industry_output``).
 - Compute ``auto_share_of_output = auto_attributed_output / total_industry_output``.
 - Output: ``sam_auto_industry_shares.csv``.
5. **Crosswalk to NAICS using IMPLAN bridges (improved)**
 - Run ``scripts/analyze_sam_auto_supply_chain.py`` first to produce industry-level results.
 - Then execute ``scripts/build_sam_auto_naics_crosswalks.py``, which:
 - Uses ``data/raw/Implan528toAggregated2022Naics.xlsx`` to create an

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(`data/intermediate/sam_naics_shares/sam_auto_naics_aggregated_shares.csv`).
- Applies `data/raw/Bridge_2022NaicsToImplan528_AllDescriptions.xlsx`
  with CEW ratios (normalized within each IMPLAN sector) to spread
  SAM industries across NAICS6, roll up to NAICS4, and align with the
  38-industry segment lookup, yielding
  `data/intermediate/sam_naics_shares/sam_auto_naics4_mobility38.csv`.
- Also saves a detailed six-digit output at
  `data/intermediate/sam_naics_shares/sam_auto_naics6_shares.csv`.
- This IMPLAN-based approach replaces the earlier BEA?NAICS
  concordance, eliminating the missing matches that arose from
  description-based joins.
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Outputs

File	Purpose
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`data/intermediate/sam_auto_commodity_shares.csv`	Commodity demand totals vs. automotive purchases
`data/intermediate/sam_auto_industry_shares.csv`	Industry-level automotive-attributed output and share
`data/intermediate/sam_auto_naics_shares.csv`	(Legacy) BEA-based NAICS view
`data/intermediate/sam_naics_shares/sam_auto_naics_aggregated_shares.csv`	IMPLAN aggregated NAICS totals
`data/intermediate/sam_naics_shares/sam_auto_naics6_shares.csv`	IMPLAN-weighted six-digit NAICS allocations
`data/intermediate/sam_naics_shares/sam_auto_naics4_mobility38.csv`	IMPLAN-derived four-digit shares aligned to mobility segments

Reproducibility

Run the script from the repository root (inside the desired Python environment):

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```bash
python scripts/analyze_sam_auto_supply_chain.py
python scripts/build_sam_auto_naics_crosswalks.py
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Optional arguments:

- `--sam-path` to override the SAM input
- `--make-table-path` to use an alternate BEA make table
- `--auto-codes` to supply a custom list of SAM industry codes
- `--output-dir` to write the outputs elsewhere